Aaryan Atul Chaudhari

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.14
Royal College of Science & Commerce, Dombivli	Class XII (Maharashtra State Board of Secondary & Higher Secondary Education)	2024	90.17%
Gurukul - The Day School	Class X (International General Certificate of Secondary Education (IGCSE))	2022	94.67%

POSITIONS OF RESPONSIBILITY

Machine Learning Core Member | Tinkerers' Laboratory, IIT Hyderabad

- **Machine Learning Orientation:** Planned and delivered an introductory machine learning orientation for **200+ students within a tight deadline**, managing content preparation and presentation delivery. Developed supporting PDF learning resources and slide presentations to explain foundational concepts, and designed short games and interactive activities to encourage participation. Confidently presented to a large student audience, combining technical explanations with engagement activities to make the domain approachable for new learners.
- **Technical Teaching -- Natural Language Processing:** Prepared and delivered an **Introduction to Natural Language Processing session for 120+ students** through the Tinker Tech initiative. Created PDF learning materials and slide presentations covering core NLP concepts and practical applications, organizing the content for an introductory audience. Delivered a well-received lecture with clear explanations and an engaging presentation style, helping students develop their understanding of NLP and its applications.

PROJECTS

Dynamo-Inspired Distributed Key-Value Store

- Built a distributed key-value store using Python, FastAPI, and Docker Compose, implementing replication, quorum operations, and write-ahead logging.
- Validated consistent hashing across **100,000** keys; expanding from three to four nodes moved **22.9%** of primary assignments, close to the theoretical 25%.
- Benchmarked **10,000**-request workloads across 10-100 concurrent clients, reaching approximately **321** requests/second at **50** clients with **100**% successful requests in the peak-throughput run.

Physics-Informed MRI Reconstruction | Python, TensorFlow, NumPy

- Built a **4-stage reconstruction pipeline** processing **~7,135 MRI slices** with **~4× undersampling**.
- Developed U-Net and **8-cascade models**, improving validation PSNR from **28.6 to 29.8 dB** and SSIM to **0.77**.
- Automated experiment tracking across **4 model variants**, achieving peak quality of **32.2 dB PSNR / 0.83 SSIM**.

Qdrant-BGE RAG Chatbot Framework | Python, Qdrant, Ollama, AWS S3, BGE-M3

- Developed a Retrieval-Augmented Generation (RAG) chatbot framework using Qdrant vector database, BGE-M3 embeddings, and Ollama LLMs for context-aware question answering.
- Implemented an end-to-end pipeline for document ingestion, semantic chunking, embedding generation, vector search, reranking, and response synthesis from user-provided AWS S3 data.
- Integrated dense retrieval, cross-encoder reranking, and conversational memory to improve retrieval relevance and maintain multi-turn context.
- Built an interactive chatbot capable of generating concise responses using retrieved knowledge while handling ambiguous or missing information robustly.

EEG-Based Alcoholism Detection using Deep Learning | Python, Tensorflow

- Processed and analyzed 11,057 EEG recordings (61 channels × 256 timepoints) from the UCI EEG dataset to classify alcoholic and control subjects.
- Implemented signal preprocessing, EEG-specific data augmentation, and MixUp regularization to improve model generalization.
- Designed and benchmarked CNN, BiLSTM, GRU, Conv1D-BiLSTM, and Conv2D-BiLSTM architectures for spatio-temporal EEG analysis.
- Achieved **94.0%** classification accuracy and demonstrated the importance of spatial feature extraction.

SKILLS

- **Soft Skills:** Leadership, Technical Communication, Public Speaking, Project Planning, Time Management
- **Programming Languages:** Python, C++, SQL
- **Libraries & Frameworks:** FastAPI, TensorFlow, scikit-learn, NumPy, Pandas
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Distributed Systems, Machine Learning
- **Tools & Platforms:** Git, GitHub, Docker Compose, AWS S3, Qdrant

RELEVANT COURSES

- **Computer Science & AI:** Data Structures and Applications, Advanced Learning Algorithms (Coursera), Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (Coursera)
- **Mathematics:** Elementary Linear Algebra, Introduction to Probability, Introduction to Statistics, Differential Equations

ACHIEVEMENTS

- Ranked **5th** in Problem Statement 1, focused on **chatbot development**, on the tentative Motiverse AI Hackathon 2025 leaderboard organized by Motive at IIT Hyderabad, qualifying among the **Top 10 teams out of approximately 150 participating teams**.

EXTRACURRICULAR

- Secured **2nd** place in Powerlifting at Diesta, IIT Hyderabad's annual interdepartmental cultural and sports festival.